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# SSM Optimization

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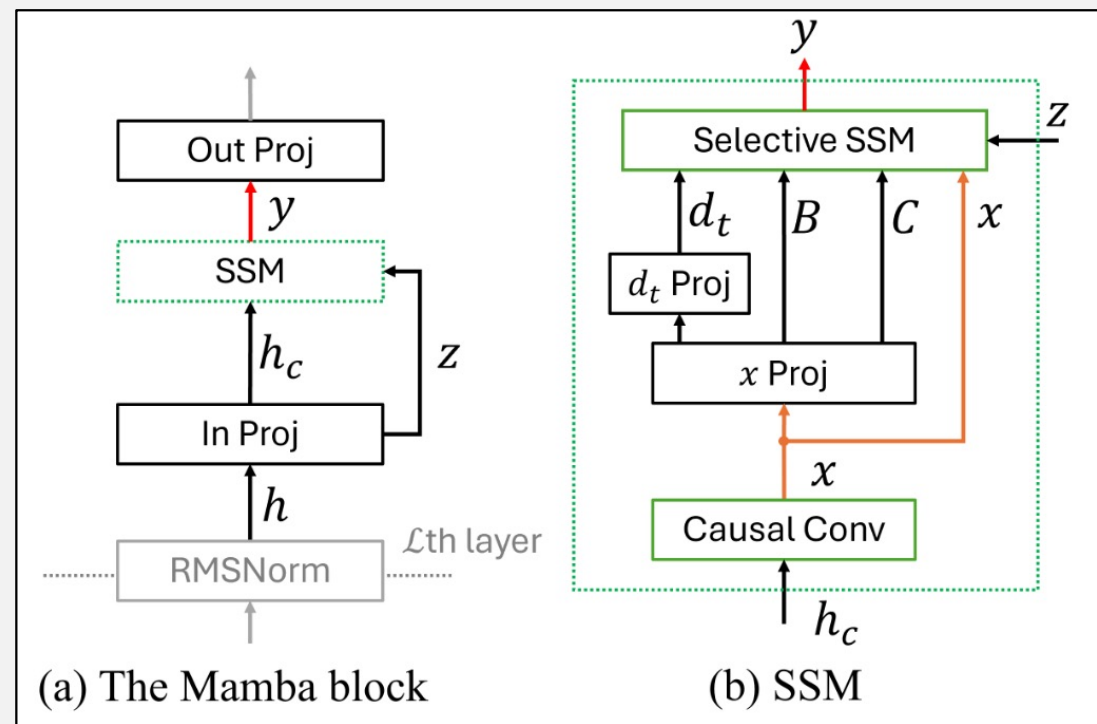
Weekly Meeting

06/03/26

Beatrice Alessandra Motetti

# Setup

- Analyzed the activations for few samples of WikiText-2 for Falcon Mamba-7B for:
  - the first Falcon Mamba block (#0);
  - the middle one (#31);
  - the last (#63).



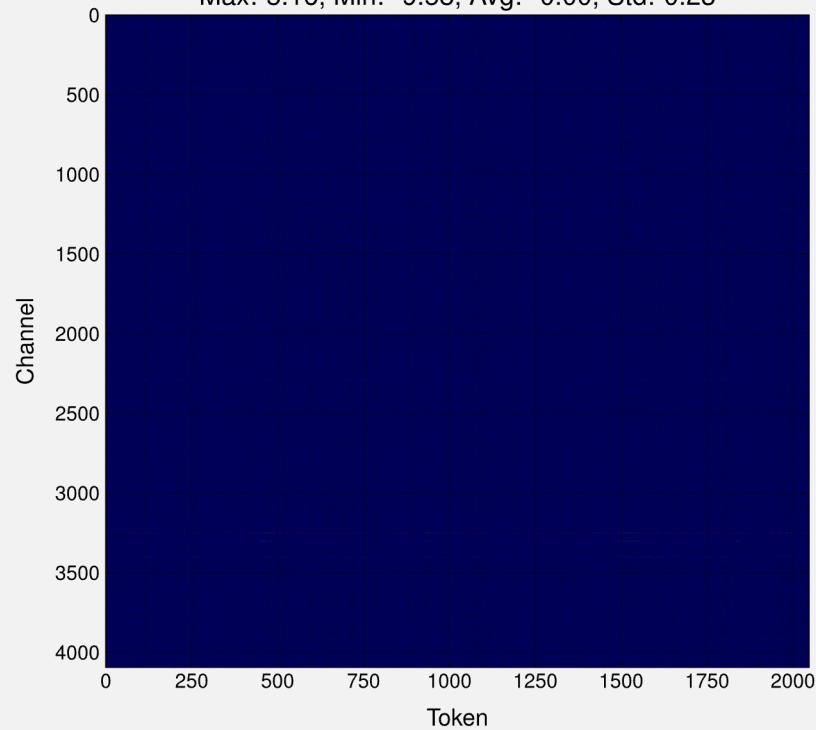
Source: Chiang, Chang et al., "Quamba: A Post-Training Quantization Recipe for Selective State Space Models", ICLR 2025

## Activations analysis

# First block: input

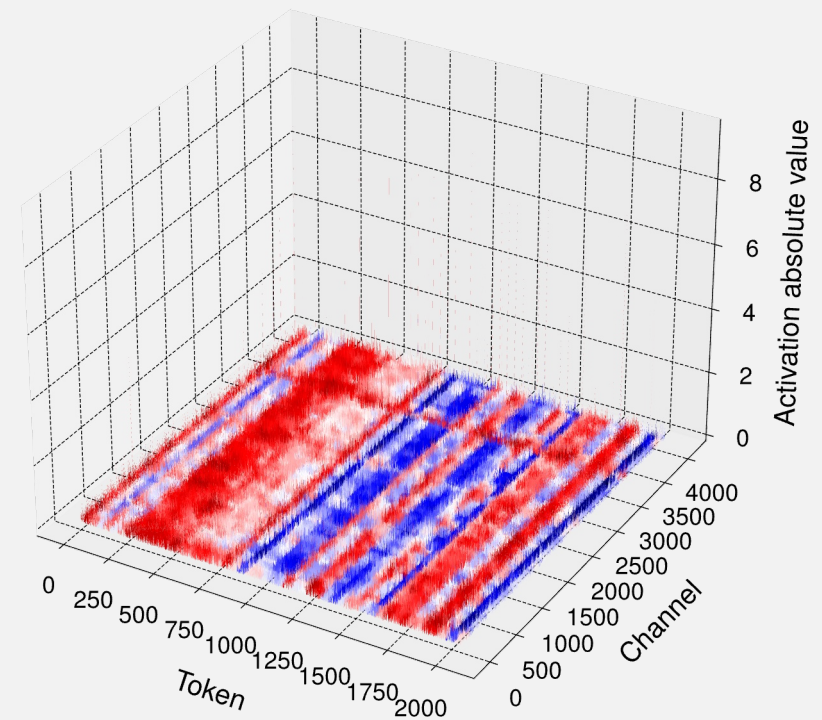
Layer: mamba\_layer\_0\_in\_proj\_input

Max: 5.16, Min: -9.58, Avg: -0.00, Std: 0.23



Layer: mamba\_layer\_0\_in\_proj\_input

Max: 5.16, Min: -9.58, Avg: -0.00, Std: 0.23

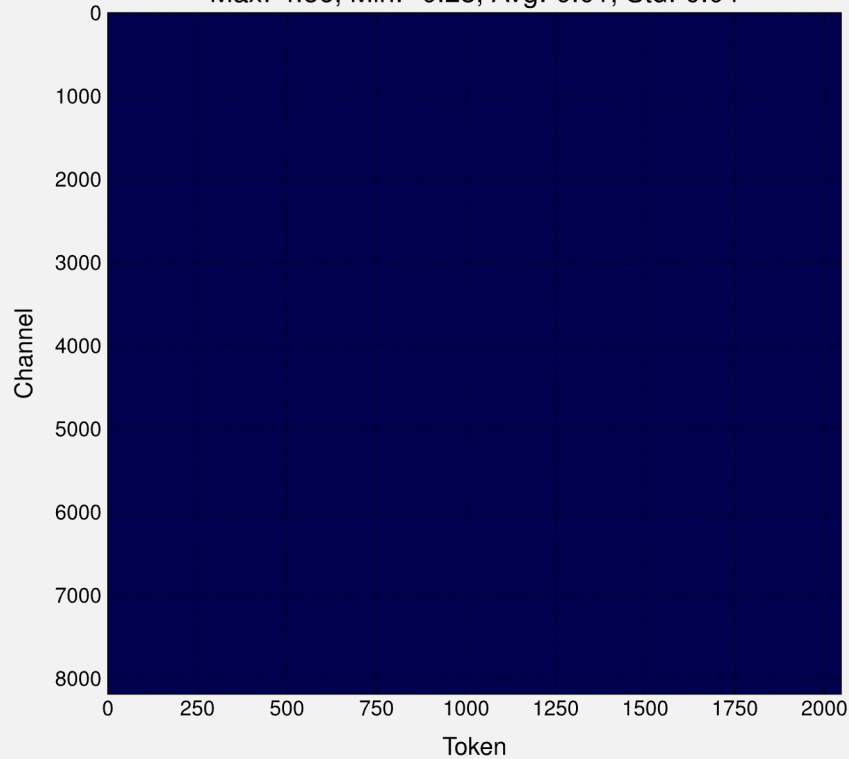


## Activations analysis

# First block: x\_proj's input

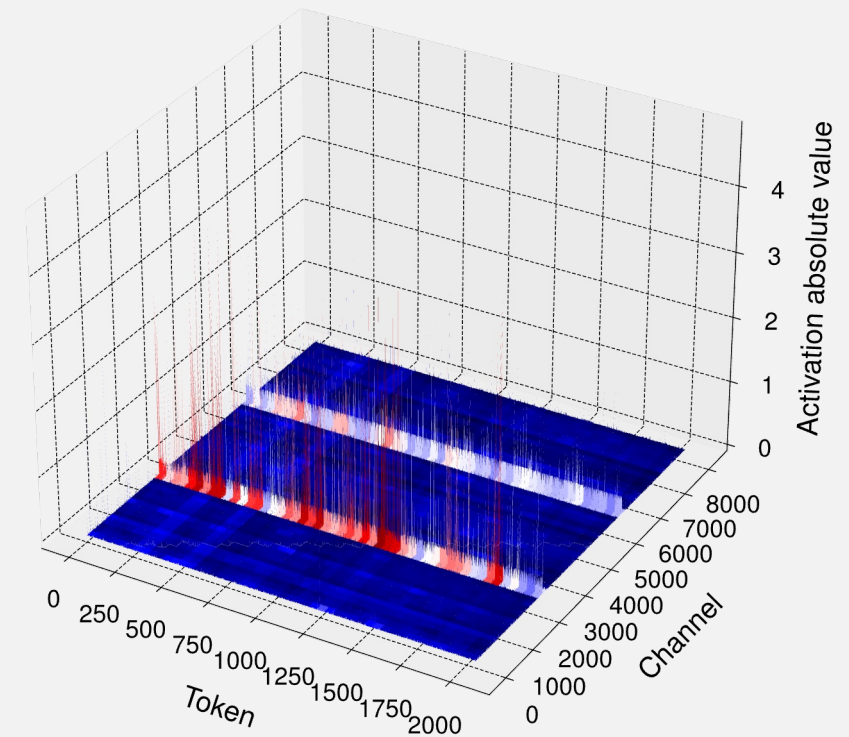
Layer: mamba\_layer\_0\_x\_proj\_input

Max: 4.86, Min: -0.28, Avg: 0.01, Std: 0.04



Layer: mamba\_layer\_0\_x\_proj\_input

Max: 4.86, Min: -0.28, Avg: 0.01, Std: 0.04



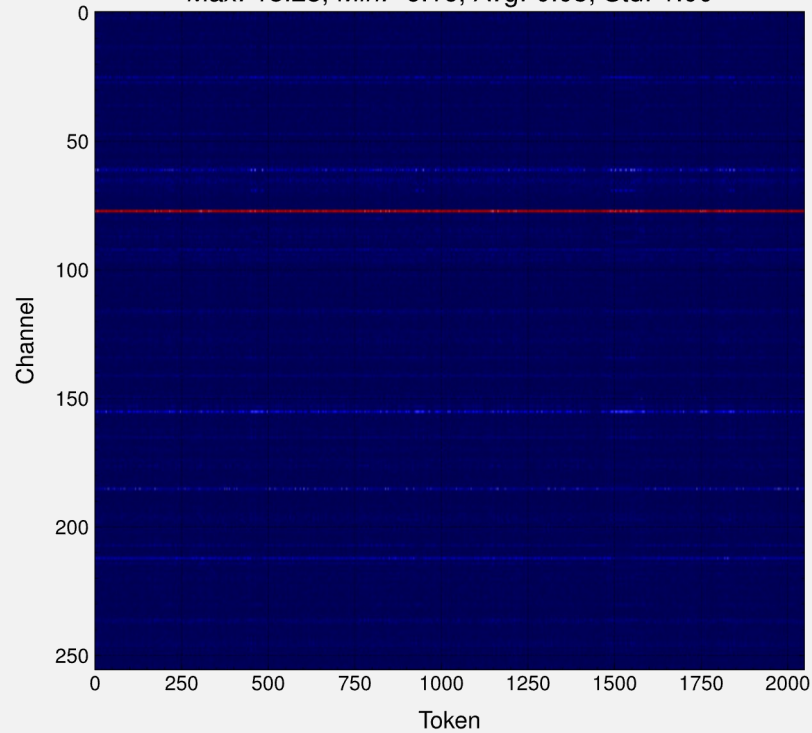


## Activations analysis

# First block: dt's input

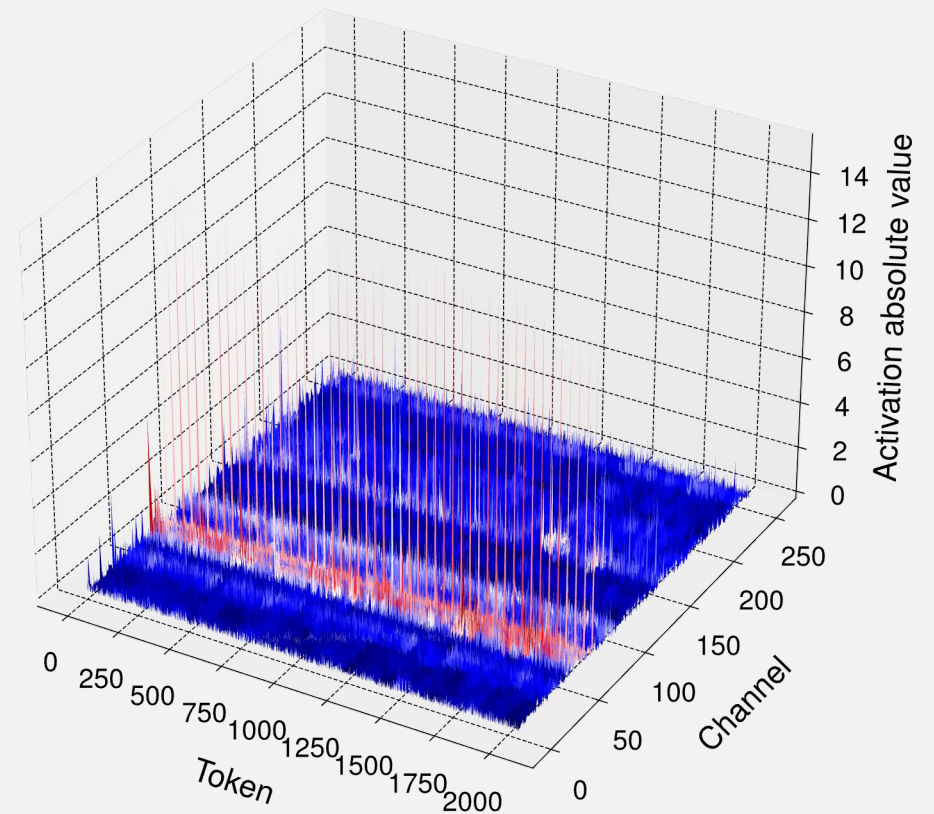
Layer: mamba\_layer\_0\_dt\_proj\_input

Max: 15.28, Min: -6.16, Avg: 0.05, Std: 1.00



Layer: mamba\_layer\_0\_dt\_proj\_input

Max: 15.28, Min: -6.16, Avg: 0.05, Std: 1.00

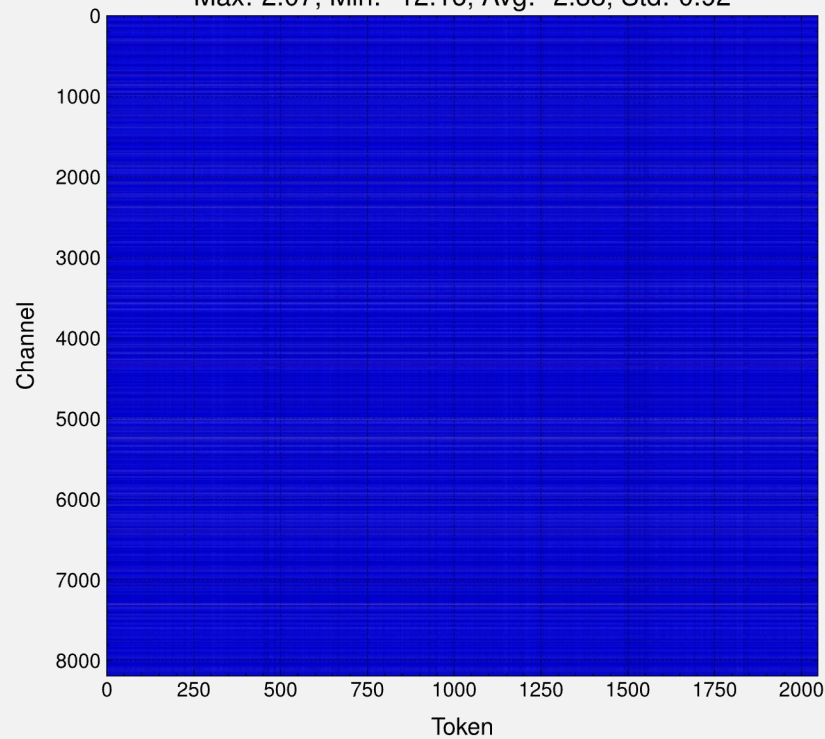


## Activations analysis

# First block: dt's output

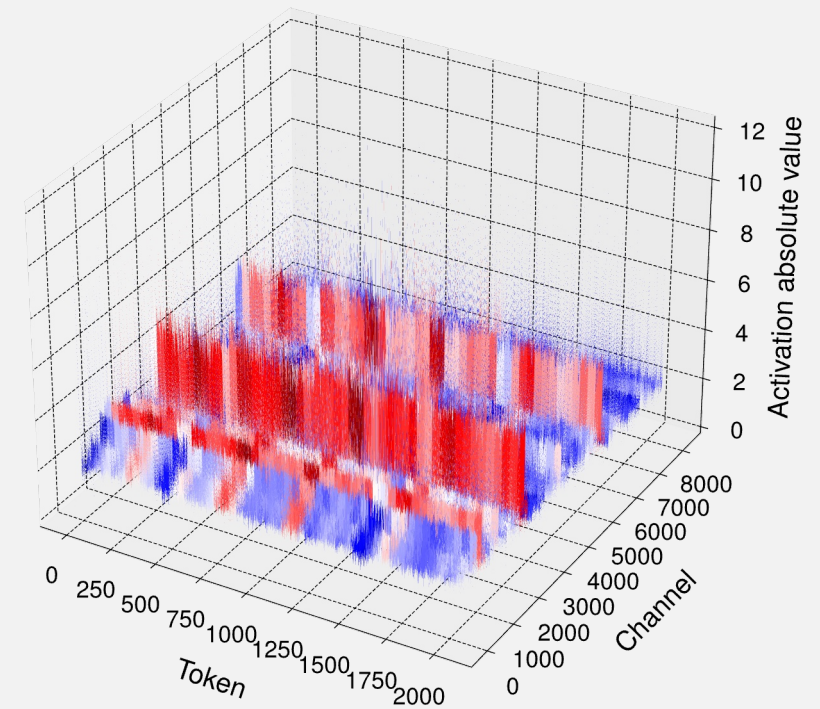
Layer: mamba\_layer\_0\_dt\_proj\_output

Max: 2.07, Min: -12.16, Avg: -2.38, Std: 0.92



Layer: mamba\_layer\_0\_dt\_proj\_output

Max: 2.07, Min: -12.16, Avg: -2.38, Std: 0.92

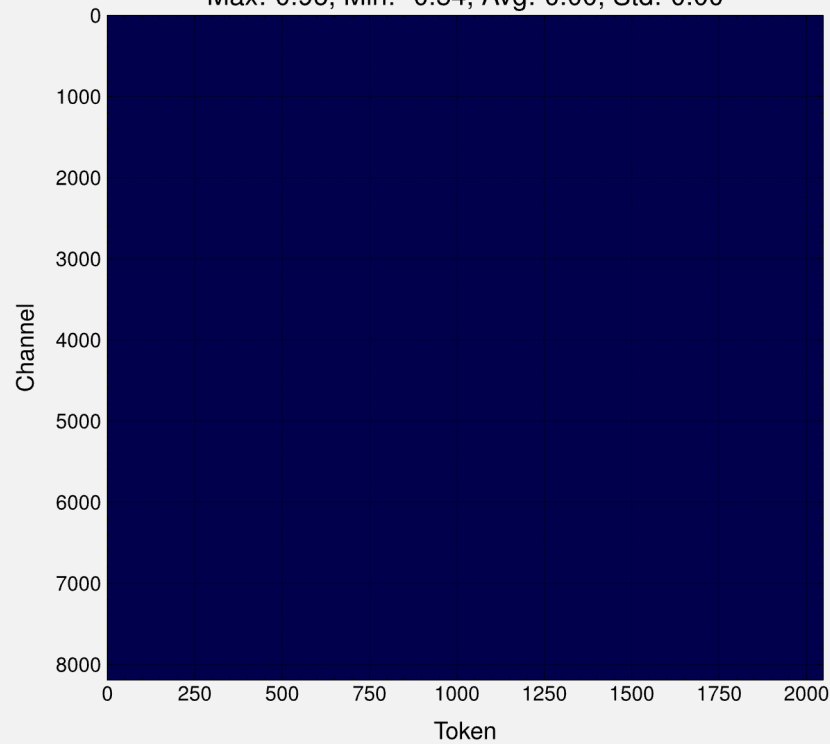


## Activations analysis

# First block: output\_proj's input

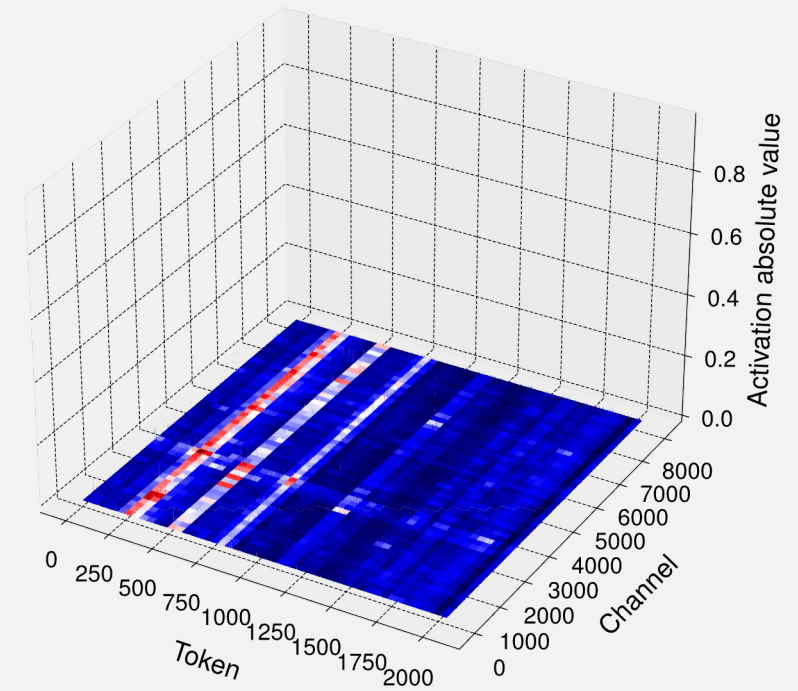
Layer: mamba\_layer\_0\_out\_proj\_input

Max: 0.96, Min: -0.54, Avg: 0.00, Std: 0.00



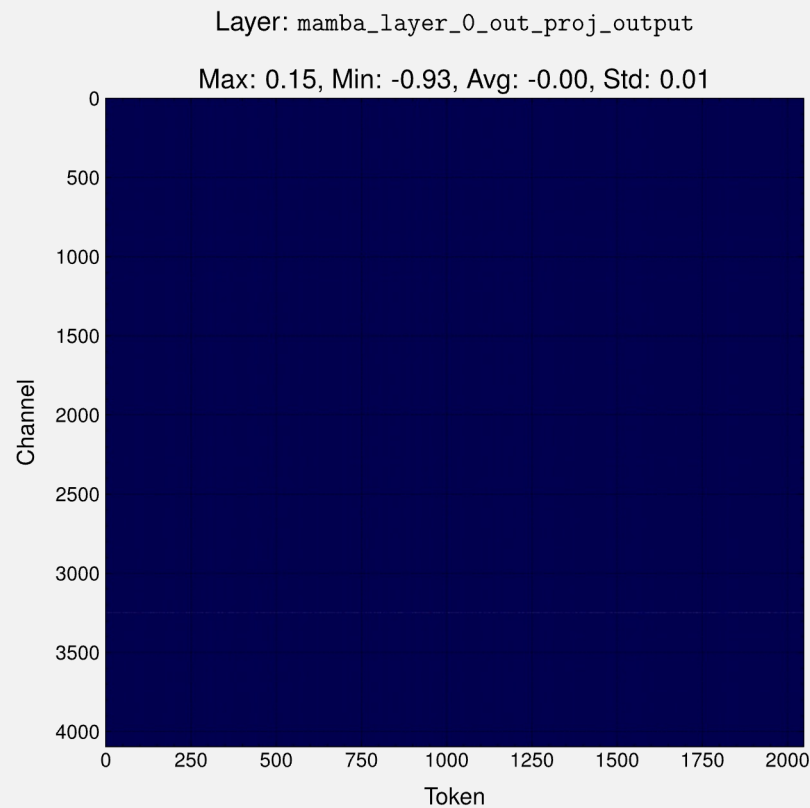
Layer: mamba\_layer\_0\_out\_proj\_input

Max: 0.96, Min: -0.54, Avg: 0.00, Std: 0.00



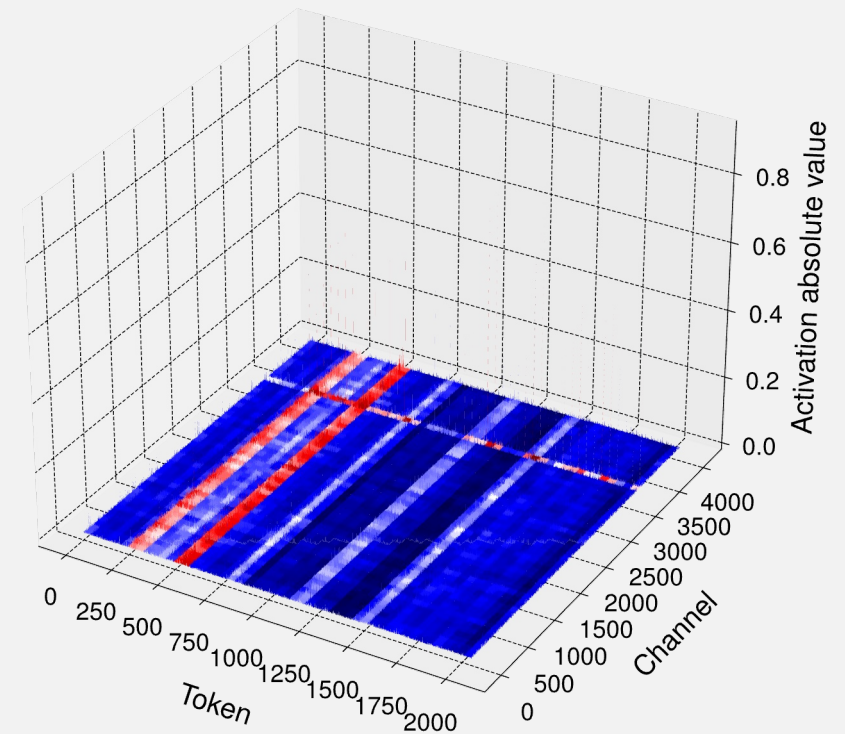
## Activations analysis

# First block: output



Layer: mamba\_layer\_0\_out\_proj\_output

Max: 0.15, Min: -0.93, Avg: -0.00, Std: 0.01



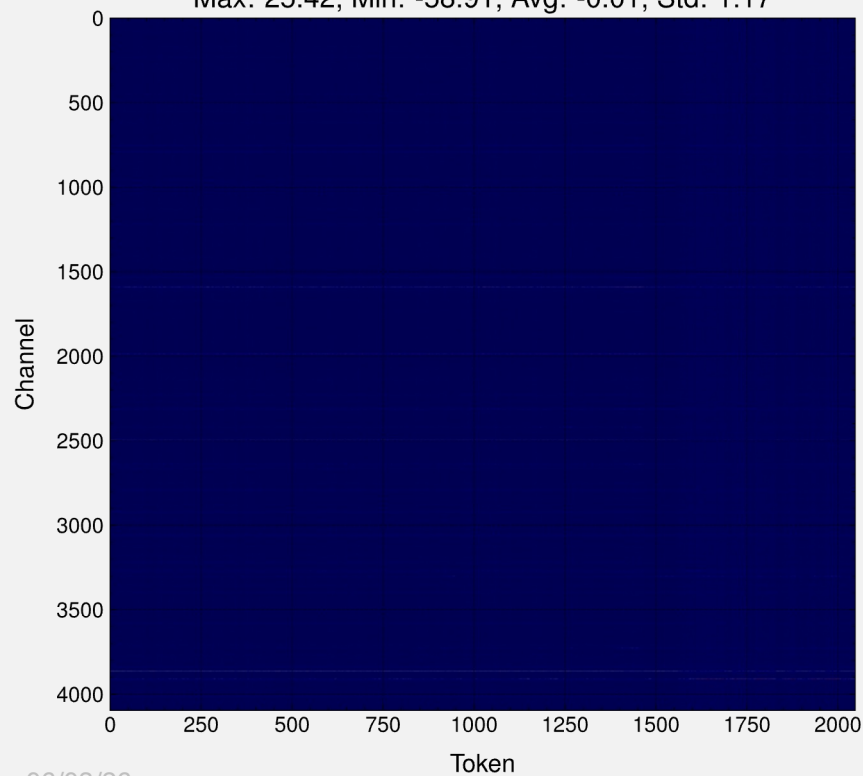


## Activations analysis

# Last block: input

Layer: mamba\_layer\_63\_in\_proj\_input

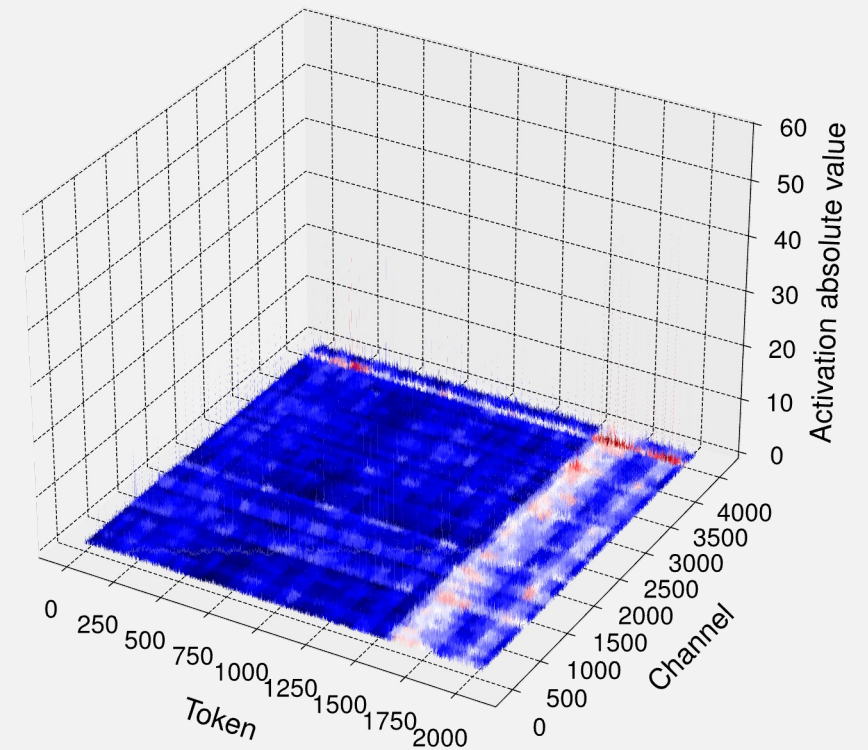
Max: 25.42, Min: -58.91, Avg: -0.01, Std: 1.17



06/03/26

Layer: mamba\_layer\_63\_in\_proj\_input

Max: 25.42, Min: -58.91, Avg: -0.01, Std: 1.17

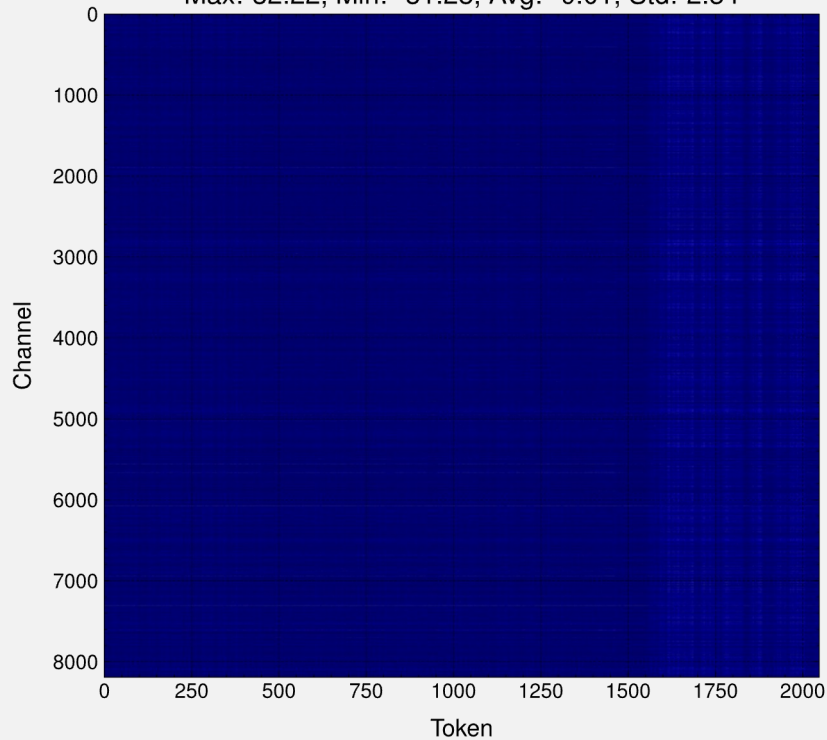


## Activations analysis

# Last block: inp\_proj's output ( $h_c$ )

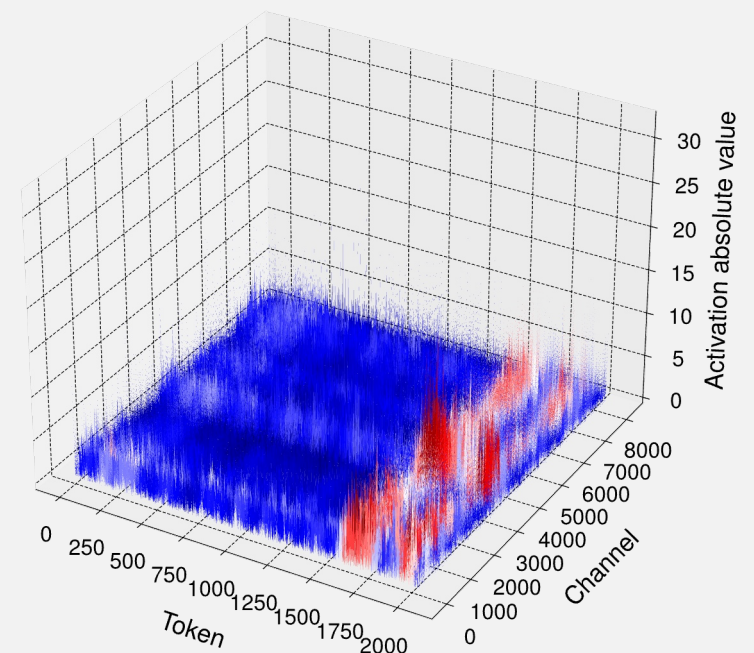
Layer: mamba\_layer\_63\_in\_proj\_hidden\_states\_hc

Max: 32.22, Min: -31.28, Avg: -0.01, Std: 2.34



Layer: mamba\_layer\_63\_in\_proj\_hidden\_states\_hc

Max: 32.22, Min: -31.28, Avg: -0.01, Std: 2.34

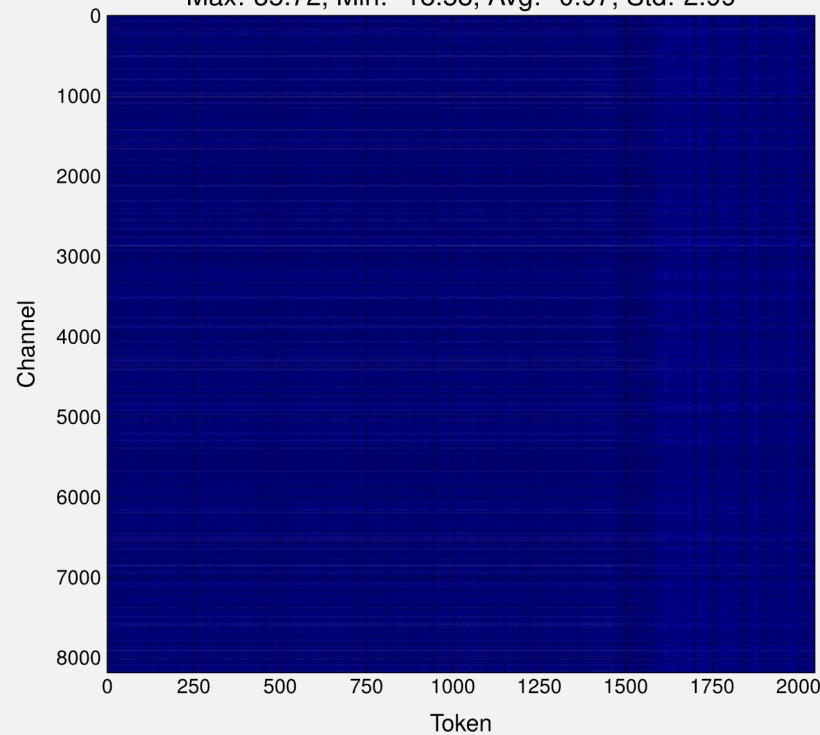


## Activations analysis

# Last block: inp\_proj's output (z)

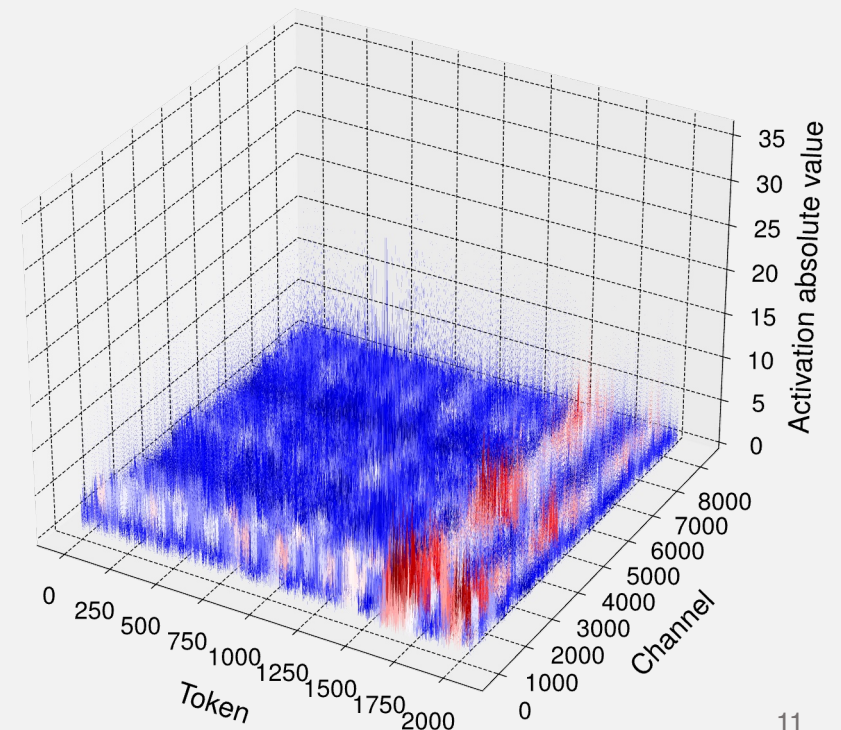
Layer: mamba\_layer\_63\_in\_proj\_gate\_z

Max: 35.72, Min: -18.58, Avg: -0.97, Std: 2.99



Layer: mamba\_layer\_63\_in\_proj\_gate\_z

Max: 35.72, Min: -18.58, Avg: -0.97, Std: 2.99

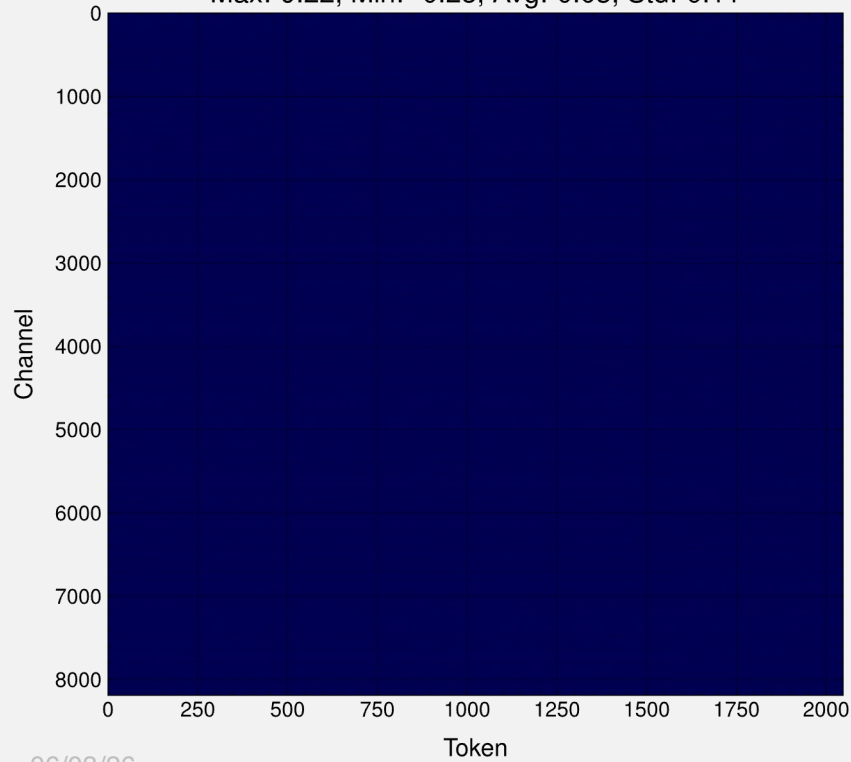


## Activations analysis

# Last block: x\_proj's input

Layer: mamba\_layer\_63\_x\_proj\_input

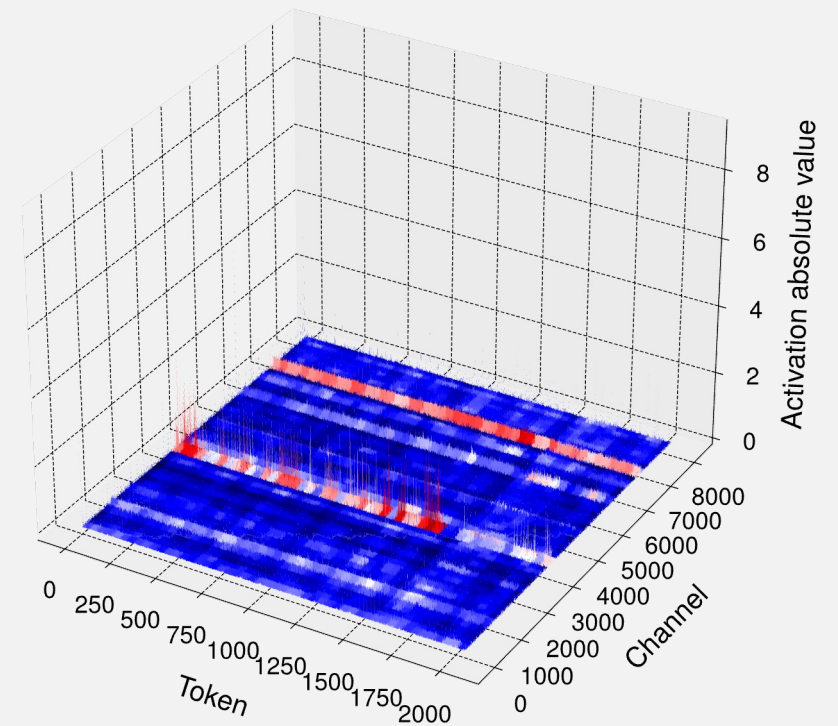
Max: 9.22, Min: -0.28, Avg: 0.03, Std: 0.11



06/03/26

Layer: mamba\_layer\_63\_x\_proj\_input

Max: 9.22, Min: -0.28, Avg: 0.03, Std: 0.11



12

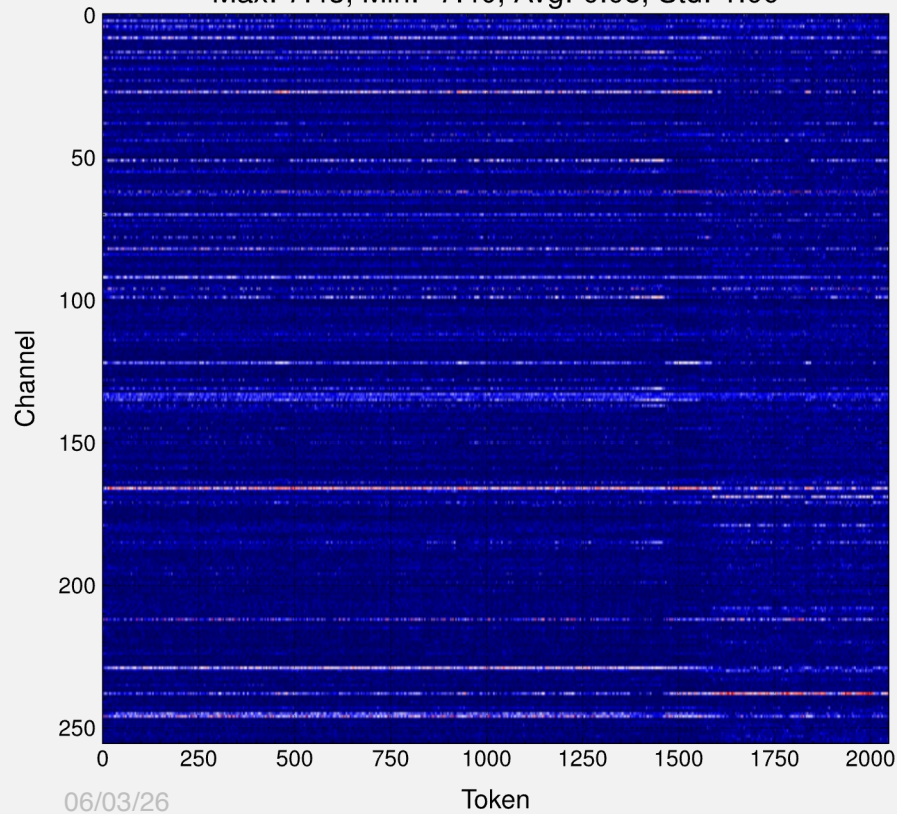


## Activations analysis

# Last block: dt's input

Layer: mamba\_layer\_63\_dt\_proj\_input

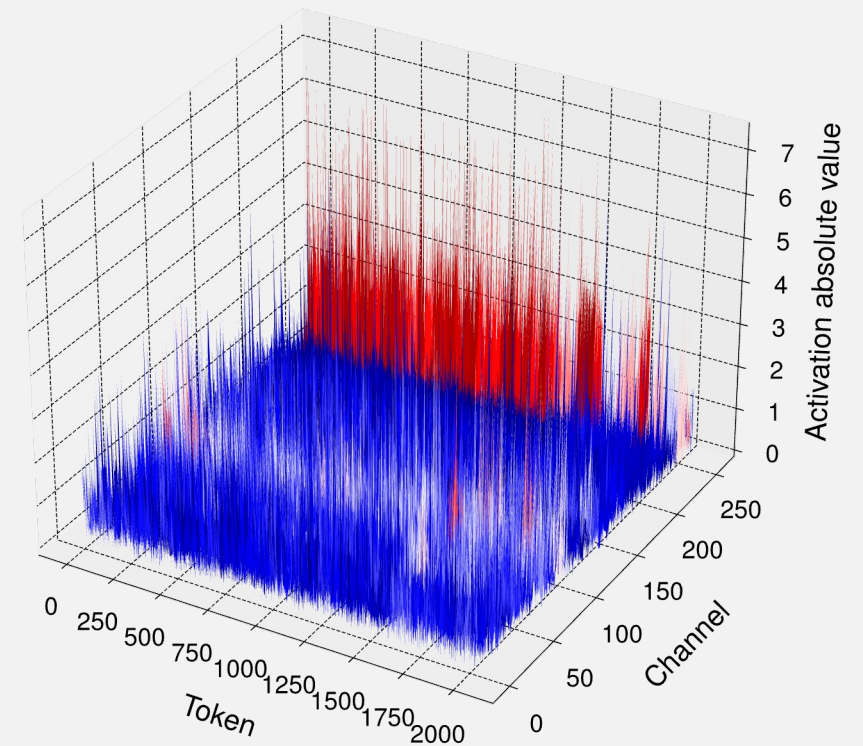
Max: 7.43, Min: -7.40, Avg: 0.08, Std: 1.00



06/03/26

Layer: mamba\_layer\_63\_dt\_proj\_input

Max: 7.43, Min: -7.40, Avg: 0.08, Std: 1.00



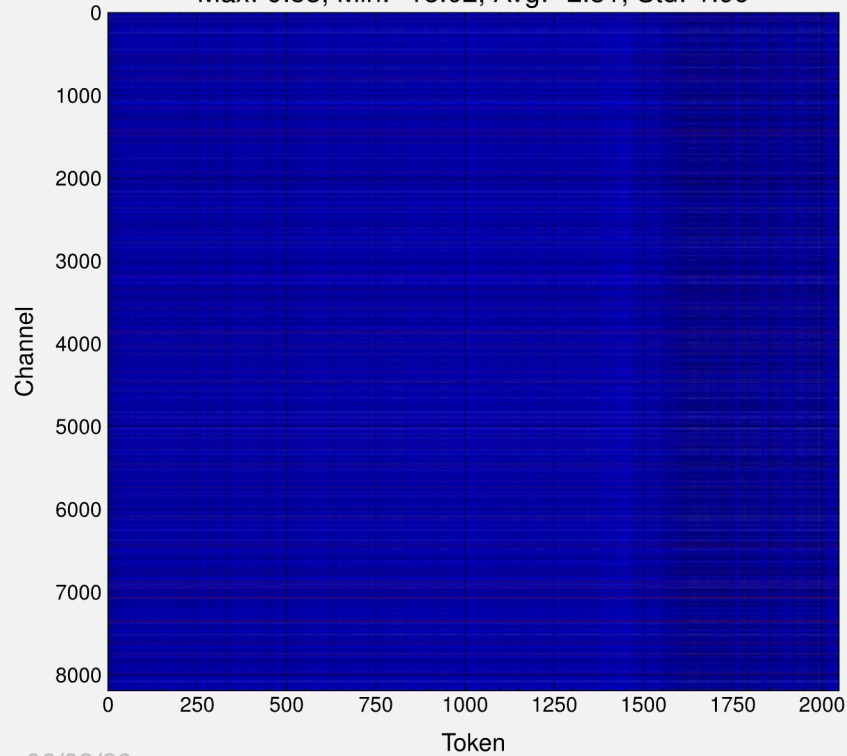
13

## Activations analysis

# Last block: dt's output

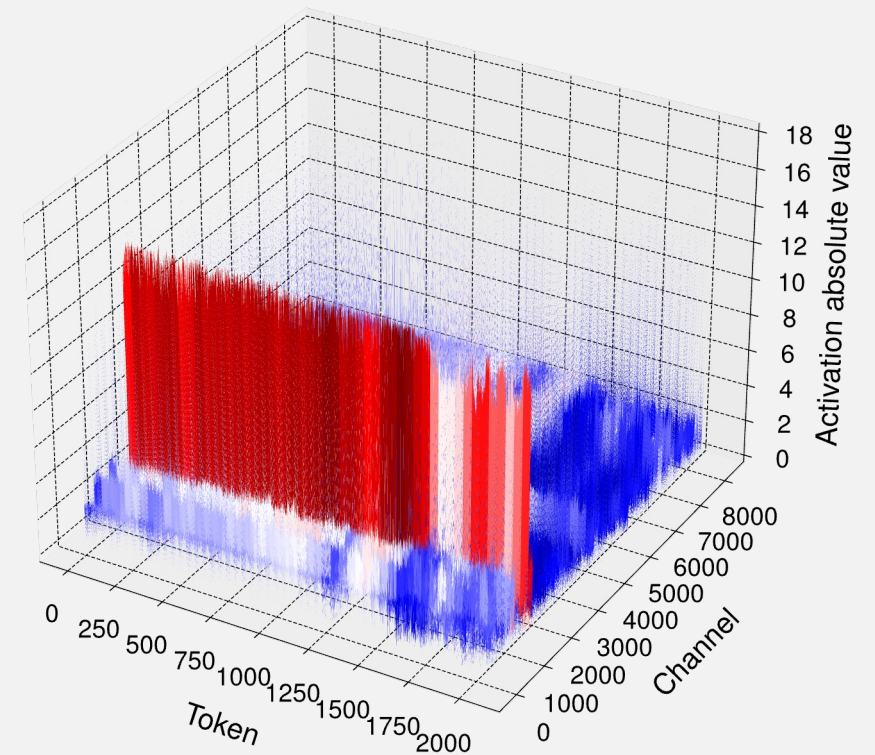
Layer: mamba\_layer\_63\_dt\_proj\_output

Max: 9.53, Min: -18.02, Avg: -2.31, Std: 1.99



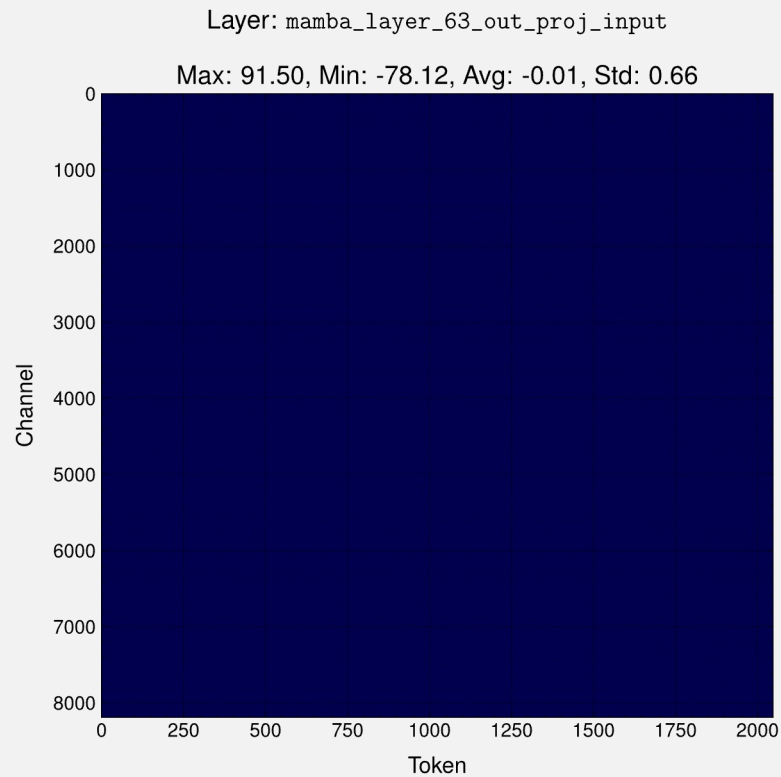
Layer: mamba\_layer\_63\_dt\_proj\_output

Max: 9.53, Min: -18.02, Avg: -2.31, Std: 1.99



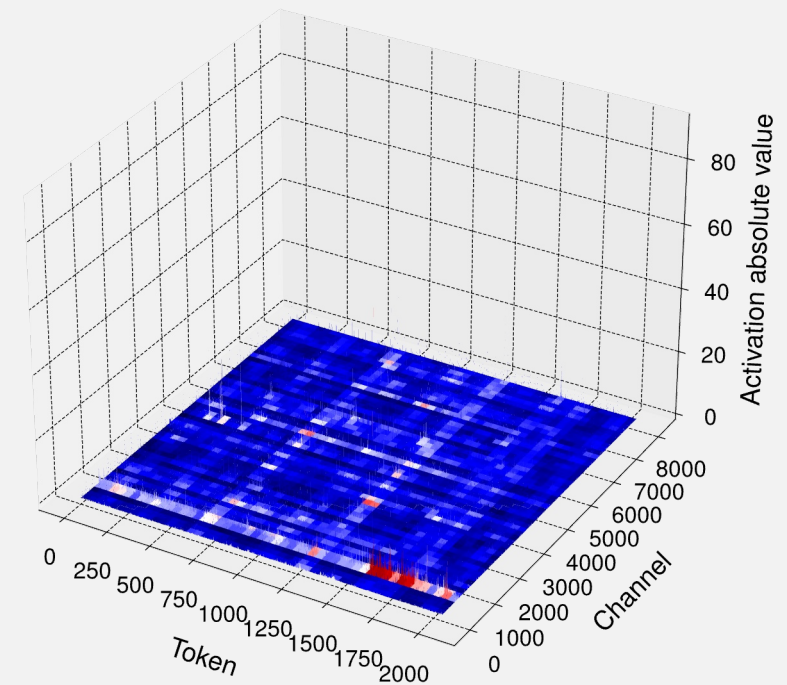
## Activations analysis

# Last block: output\_proj's input



Layer: mamba\_layer\_63\_out\_proj\_input

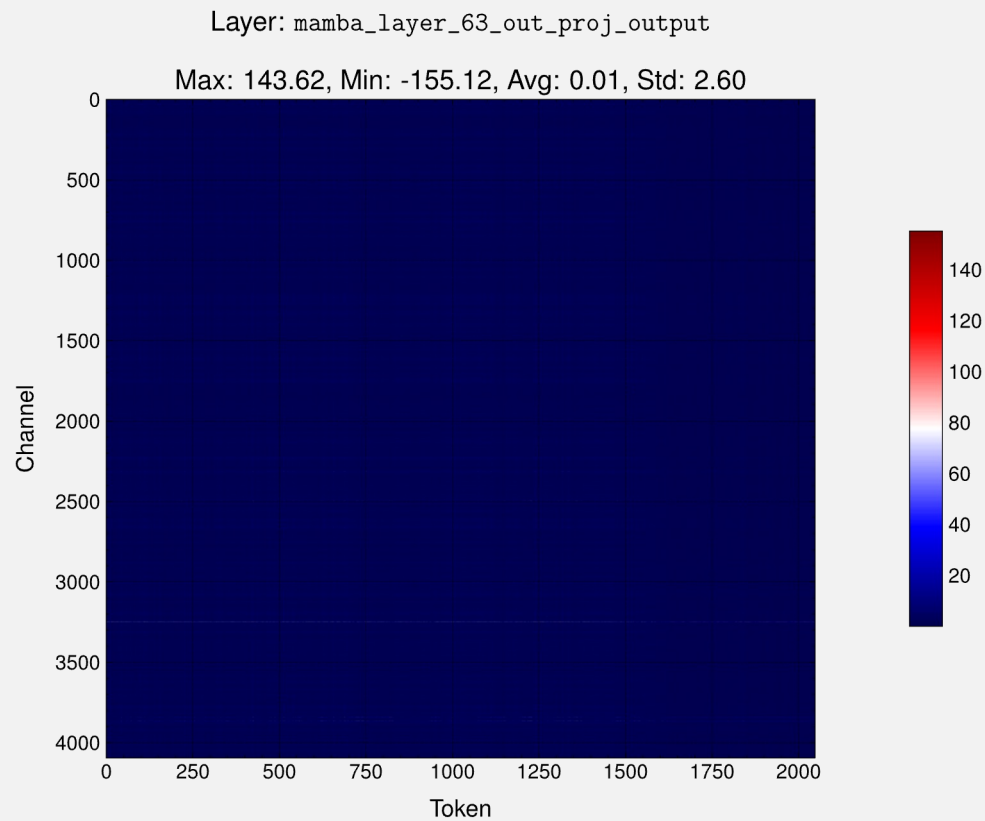
Max: 91.50, Min: -78.12, Avg: -0.01, Std: 0.66





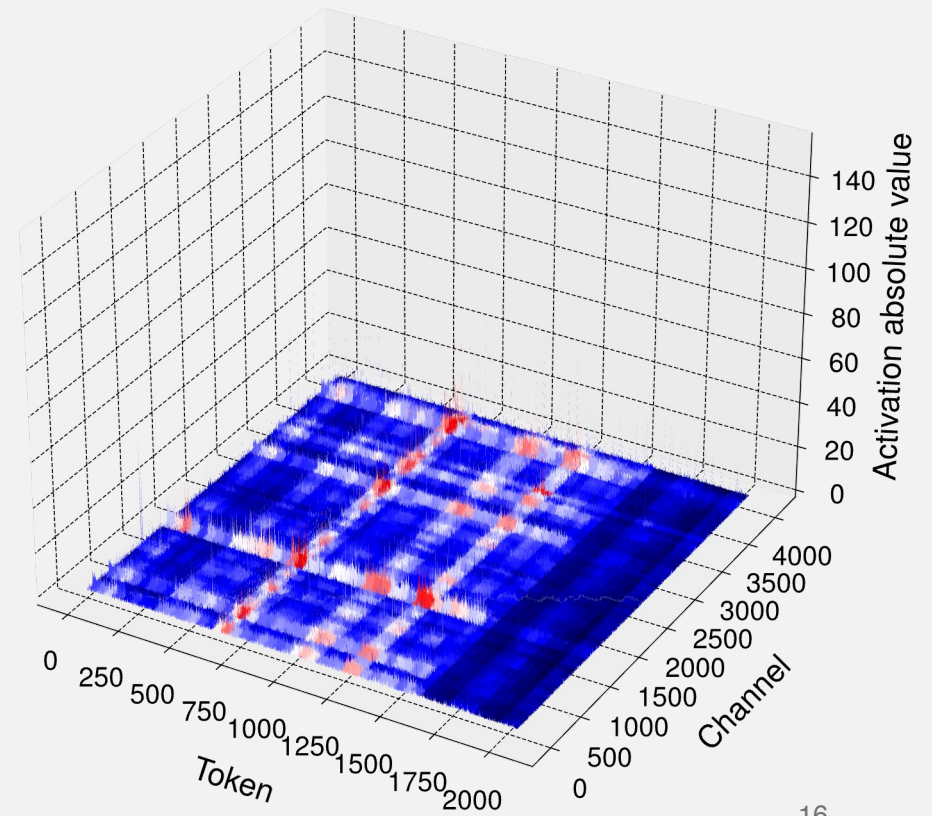
## Activations analysis

# Last block: output



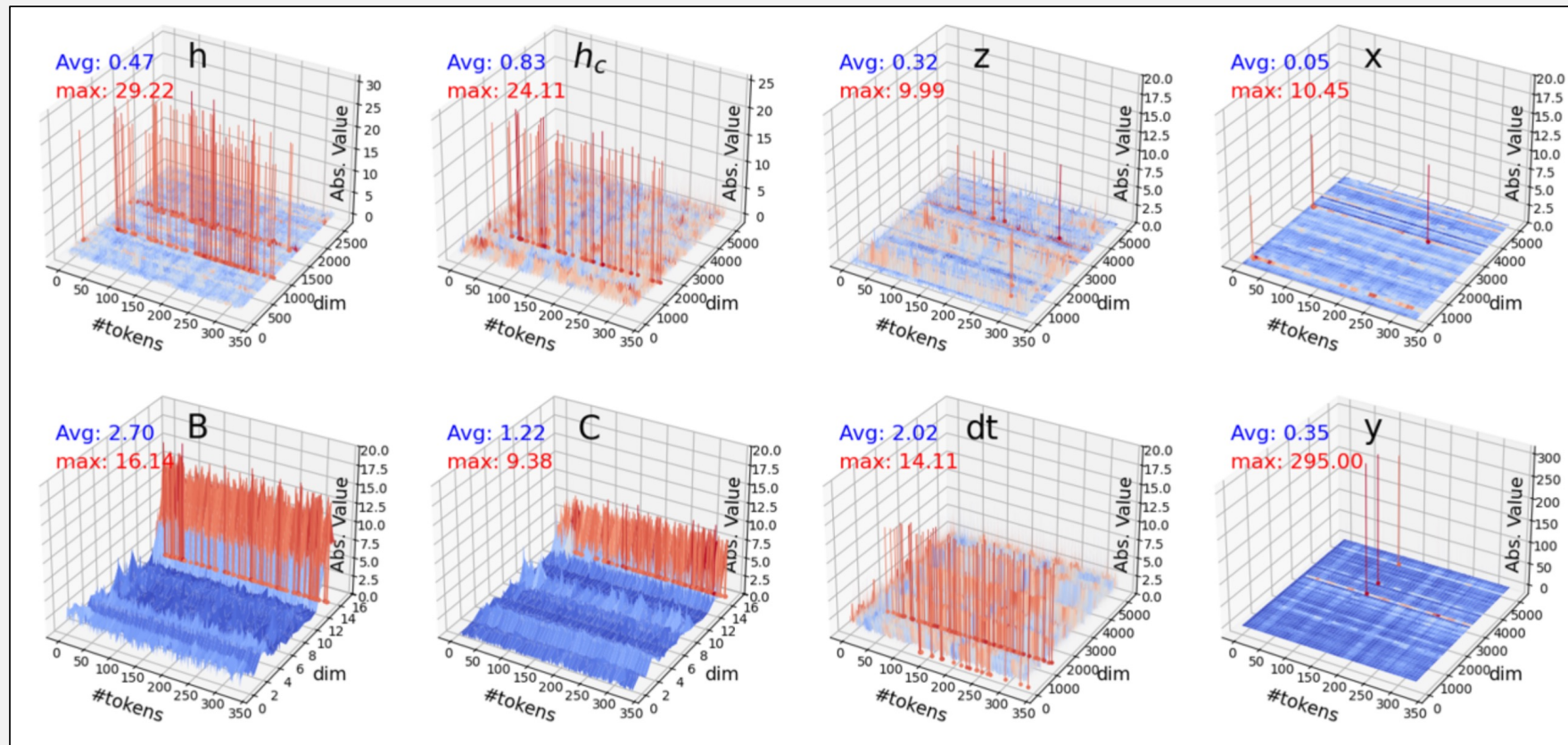
Layer: mamba\_layer\_63\_out\_proj\_output

Max: 143.62, Min: -155.12, Avg: 0.01, Std: 2.60



## Activations analysis

# Comparison with Quamba





# Benchmark

Table 2: Comparative results under different quantization settings on language mamba models. Evaluations on the five standard datasets—ARC-E, ARC-C, PIQA, Winogrande and HellaSwag—resulted in the reported average accuracy across these datasets.

| Bit Width | Methods     | Mamba-LLM  |            |            |            |
|-----------|-------------|------------|------------|------------|------------|
|           |             | Mamba-370m | Mamba-790m | Mamba-1.4b | Mamba-2.8b |
| FP16      | -           | 50.9       | 54.8       | 58.6       | 62.2       |
| W8A8      | RTN         | 45.7       | 44.9       | 53.9       | 58.4       |
|           | GPTQ+RTN    | 46.2       | 48.6       | 55.0       | 58.9       |
|           | SmoothQuant | 45.2       | 41.7       | 54.2       | 58.7       |
|           | QuaRot      | 48.8       | 51.6       | 56.9       | 59.3       |
|           | Ours        | 50.0       | 53.8       | 58.3       | 62.1       |
| W4A8      | RTN         | 36.2       | 35.4       | 51.6       | 54.8       |
|           | GPTQ+RTN    | 36.7       | 36.0       | 51.1       | 53.6       |
|           | SmoothQuant | 36.8       | 39.3       | 52.0       | 54.9       |
|           | QuaRot      | 43.4       | 40.0       | 53.8       | 58.5       |
|           | Ours        | 43.9       | 45.8       | 54.3       | 58.5       |

Source: Xu et al., "MambaQuant: Quantizing the Mamba Family with Variance Aligned Rotation Methods", ICLR 2025

## Comparison with baselines

# Pt. 1

| Model           | Quantized  | Wikitext2  |               |            |               |              |               |            |               |
|-----------------|------------|------------|---------------|------------|---------------|--------------|---------------|------------|---------------|
|                 |            | SINQ       |               | RTN        |               | bitsandbytes |               | HQQ        |               |
|                 |            | Perplexity | KL divergence | Perplexity | KL divergence | Perplexity   | KL divergence | Perplexity | KL divergence |
| Falcon Mamba 7B | 🚫          | 5.203      | -             | -          | -             | -            | -             | -          | -             |
| Falcon Mamba 7B | ✅ (8 bits) | 5.159      | 18.946        | 5.158      | 18.965        | 16050115.0   | 30605.516     | 5.159      | 0.858         |
| Falcon Mamba 7B | ✅ (4 bits) | 5.378      | 99.482        | 5.397      | 103.422       | 5.591        | 182.742       | 5.365      | 78.783        |
| Falcon Mamba 7B | ✅ (3 bits) | 5.987      | 331.053       | 6.035      | 349.105       | -            | -             | 5.932      | 291.072       |
| Falcon Mamba 7B | ✅ (2 bits) | 18.514     | 2633.312      | 36.174     | 3995.665      | -            | -             | 11.798     | 1704.522      |
| Qwen3-8B        | 🚫          | 9.727      | -             | -          | -             |              |               |            |               |
| Qwen3-8B        | ✅ (8 bits) | 9.716      | 1.778         | 9.719      | 2.056         |              |               |            |               |
| Qwen3-8B        | ✅ (4 bits) | 9.838      | 127.373       | 9.974      | 160.756       |              |               | 9.798      | 159.876       |
| Qwen3-8B        | ✅ (3 bits) | 11.749     | 579.109       | 12.391     | 712.955       | -            | -             |            |               |
| Qwen3-8B        | ✅ (2 bits) | 8573.92    | 14971.098     | 14965.382  | 16119.065     | -            | -             |            |               |

## Comparison with baselines

# Pt. 2

| Model      | Quantized  | Wikitext2  |               |            |               |              |               |            |               |
|------------|------------|------------|---------------|------------|---------------|--------------|---------------|------------|---------------|
|            |            | SINQ       |               | RTN        |               | bitsandbytes |               | HQQ        |               |
|            |            | Perplexity | KL divergence | Perplexity | KL divergence | Perplexity   | KL divergence | Perplexity | KL divergence |
| Mamba-1.4B | 🚫          | 10.761     | -             | -          | -             | -            | -             | -          | -             |
| Mamba-1.4B | ✅ (8 bits) | 10.753     | 1.331         | 10.753     | 1.266         | NaN          | NaN           |            |               |
| Mamba-1.4B | ✅ (4 bits) | 11.094     | 64.104        | 11.225     | 79.748        | 125907.5625  | 19254.400     | 11.172     | 72.516        |
| Mamba-1.4B | ✅ (3 bits) | 12.627     | 326.925       | 12.870     | 375.701       | -            | -             |            |               |
| Mamba-1.4B | ✅ (2 bits) | 64.315     | 3626.683      | 88.682     | 4263.727      | -            | -             |            |               |
| Qwen3-1.7B | 🚫          | 16.714     | -             | -          | -             | -            | -             | -          | -             |
| Qwen3-1.7B | ✅ (8 bits) | 16.645     | 2.704         | 16.66      | 4.22          | 3990734.25   | 26157.589     |            |               |
| Qwen3-1.7B | ✅ (4 bits) | 17.189     | 229.498       | 18.75      | 375.382       | 24.042       | 654.945       | 18.964     | 420.080       |
| Qwen3-1.7B | ✅ (3 bits) | 23.268     | 1107.034      | 32.498     | 1814.610      | -            | -             |            |               |
| Qwen3-1.7B | ✅ (2 bits) | 25379.449  | 16381.184     | 202540.781 | 21139.056     | -            | -             |            |               |

## Comparison with baselines

# Pt. 3

| Model      | Quantized  | Wikitext2  |               |            |               |              |               |            |               |
|------------|------------|------------|---------------|------------|---------------|--------------|---------------|------------|---------------|
|            |            | SINQ       |               | RTN        |               | bitsandbytes |               | HQQ        |               |
|            |            | Perplexity | KL divergence | Perplexity | KL divergence | Perplexity   | KL divergence | Perplexity | KL divergence |
| Mamba-2.8B | 🚫          | 9.462      | -             | -          | -             | -            | -             | -          | -             |
| Mamba-2.8B | ✅ (8 bits) | 9.450      | 3.230         | 9.450      | 3.228         | NaN          | NaN           | 9.454      | 0.273         |
| Mamba-2.8B | ✅ (4 bits) | 9.713      | 61.813        | 9.724      | 65.213        | 613.784      | 8598.400      | 9.717      | 56.383        |
| Mamba-2.8B | ✅ (3 bits) | 10.818     | 302.866       | 10.822     | 295.758       | -            | -             |            |               |
| Mamba-2.8B | ✅ (2 bits) | 72.718     | 4181.002      | 65.736     | 3985.892      | -            | -             |            |               |
| Qwen3-4B   | 🚫          | 13.666     | -             | -          | -             |              |               |            |               |
| Qwen3-4B   | ✅ (8 bits) | 13.628     | 2.109         | 13.623     | 2.301         | 2773487.5    | 26389.918     |            |               |
| Qwen3-4B   | ✅ (4 bits) | 14.721     | 195.475       | 14.908     | 224.608       | 1086578.375  | 24838.137     | 15.706     | 243.180       |
| Qwen3-4B   | ✅ (3 bits) | 19.220     | 959.884       | 19.311     | 1088.881      | -            | -             |            |               |
| Qwen3-4B   | ✅ (2 bits) | 72940.484  | 18836.552     | 82132.156  | 19461.636     | -            | -             |            |               |



## Comparison with baselines

# bitsandbytes – debug at 8 bits

(Qwen3-0.6B)

```
(Pdb) tokenizer.decode(batch)
```

```
[' have produced creditable poems by his early teens , but these have been lost . \n\n\n In the early 730s , he travelled in the Jiang su / Zhejiang area ; his earliest surviving poem , describing a poetry contest , is thought to date from the end of this period , arou nd 735 . In that year , he took the civil service exam , likely in Chang \\'an . He failed , to his surprise and that of centuries of l ater critics . Hung concludes that he probably failed because his prose style at the time was too dense and obscure , while Chou sugge sts his failure to cultivate connections in the capital may have been to blame . After this failure , he went back to traveling , this time around Shandong and Hebei . \n\n\n His father died around 740 . Du Fu would have been allowed to enter the civil service because of his father \\'s rank , but he is thought to have given up the privilege in favour of one of his half brothers . He spent the next f our years living in the Luoyang area , fulfilling his duties in domestic affairs . \n\n\n In the autumn of 744 , he met Li Bai ( Li Po
```

```
(Pdb) tokenizer.decode(pred_fp)
```

```
['licants a to results, the own years, and he poems not mostly to The** the poem years10ss, the was to the northnan region Zhejiang re gion, he work known poem is the the scene competition in is the to be to the of the period . and 730 . He the period , he wrote part first service examination and and to the\'ans , He was the but which great , surprise of his later tradition generations ,\n\n\n \' t hat the was did because of poetry was was that time was too formal and his , and hisao says that failure was be his with the society c ity have been the his .\n\n\n the period , he continued to to the in and time in theenzhenong , intobei , He The later died in 740 , H e Fu died have been born to take the imperial service examination of his father\'ss death and but he failed not to have been up his se rvice and the of a of his own-s . He was his next years years in in the countrysideoyang region , and his role as the affairs and He I n the early of 745 , he took his Bai and Li Bai ), , the first time . and they two became met a friendship . They H , this meeting a a
```

```
(Pdb) tokenizer.decode(pred_q)
```

```
['math\ createSelectoratory...aque ,.....{. 25..., in, 2.,hei..; (hest...,hestt,utva to from-hot end of.호,호 7 5 ,va,,北方,.. ,In Chang \'. .ut, to error and ,\n\n,\n 0,_bn,호 error, error_bn, is,_bn, too, and absent,..acro ( failure to_bn, in호 mate... to. ,,-bTooto,..호 andapa and..... parent.中央 52.... would._b, to,,_bn. there, ( parentets.5 ## but,,臀 to,acro to, eggs In_ac,. of (,古 .bs中央. four..In..臀 in,-b ( in ..... 155,, if.. (... for. first., and.호的古 . . ham,. " mosty4.,pl in.. \':酸 edly " because it. import. example of., 早...的edly to., was麻痹after ( failure in,... :: is还,ント..隆 however .. wasarat +,宗 w hile.hes古古 古. We. 墓 to aboutGreatGreat from 早. butdepeenth and宗 other. They臀 again once in.115 ...111 老老 to remoteaste in.. toedly古古堂革命古. almost호.. second.osal>To+, but all almostav isav古 almost神祖av@iningly古 remote almost隆 of速edly _ac1 again古 almost古,古.ing solar호.在 15. , 151 and probablyAgain in 155 1著 yaklaşık 152 , and古 151 edly had had 1.老宗 and two호, ..
```

## Comparison with baselines

# bitsandbytes – debug at 4 bits

(Qwen3-0.6B)

```
(Pdb) tokenizer.decode(batch)
```

```
[' have produced creditable poems by his early teens , but these have been lost . \n\n\n In the early 730s , he travelled in the Jiang su / Zhejiang area ; his earliest surviving poem , describing a poetry contest , is thought to date from the end of this period , arou nd 735 . In that year , he took the civil service exam , likely in Chang \\'an . He failed , to his surprise and that of centuries of l ater critics . Hung concludes that he probably failed because his prose style at the time was too dense and obscure , while Chou sugge sts his failure to cultivate connections in the capital may have been to blame . After this failure , he went back to traveling , this time around Shandong and Hebei . \n\n\n His father died around 740 . Du Fu would have been allowed to enter the civil service because of his father \\'s rank , but he is thought to have given up the privilege in favour of one of his half brothers . He spent the next f our years living in the Luoyang area , fulfilling his duties in domestic affairs . \n\n\n In the autumn of 744 , he met Li Bai ( Li Po
```

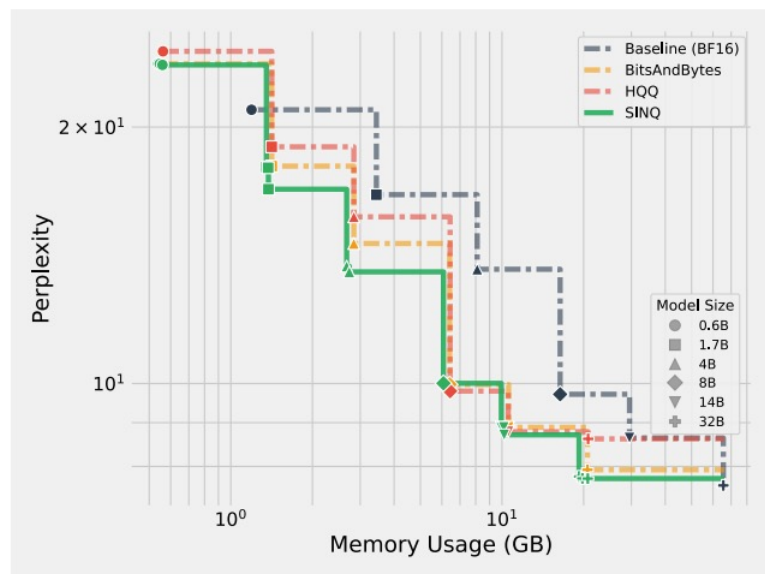
```
(Pdb) tokenizer.decode(pred_fp)
```

```
['licants a to results, the own years, and he poems not mostly to The** the poem years10ss, the was to the northnan region Zhejiang re gion , he work known poem is the the scene competition in is the to be to the of the period . and 730 . He the period , he wrote part first service examination and and to the\'ans , He was the but which great , surprise of his later tradition generations ,\n\n\n \' t hat the was did because of poetry was was that time was too formal and his , and hisao says that failure was be his with the society c ity have been the his .\n\n\n the period , he continued to to the in and time in theenzhenong , intobei , He The later died in 740 , H e Fu died have been born to take the imperial service examination of his father\'ss death and but he failed not to have been up his se rvice and the of a of his own-s . He was his next years years in in the countrysideoyang region , and his role as the affairs and He I n the early of 745 , he took his Bai and Li Bai ) , , the first time . and they two became met a friendship . They H , this meeting a a
```

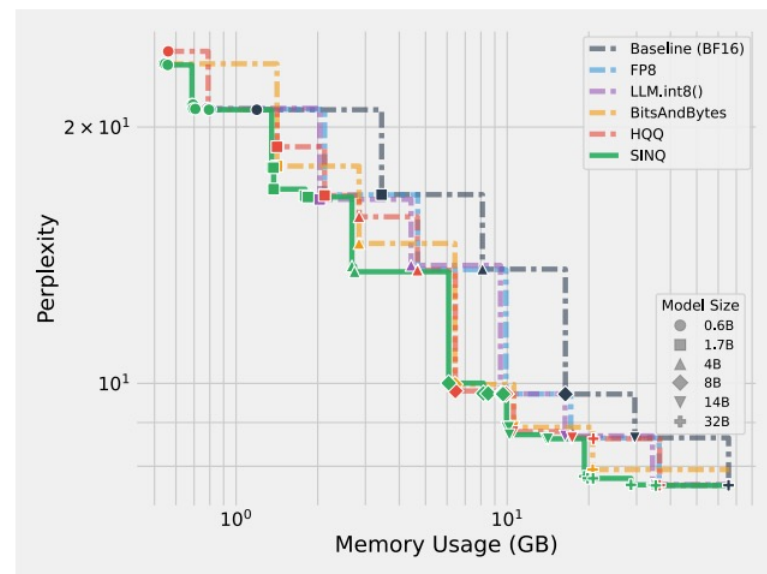
```
(Pdb) tokenizer.decode(pred_q)
```

```
[' a to results, the own years, and he are not in to The** the end years100s, the was to the eastnan region Hhejiang region and and f amily known poem is " the scene competition in is the to be to the of the period. and 150- \n\n the period , he wrote a first service examination in and to the\'ans or He was in but be great , disappointment is his later Chinese generations and \n\nry that the is wro te to of own was was that time was too much and his in but hisao\'s that failure was meet his to the Chinese city be been the his . \n \n the period , he returned to to the in and time to ,uong , thebei , He The name was in 150 , He Fu died have been born to go the imp erial service examination of his father\'ss death and but he was not to have died up his role of his of his of his children-s . This d ied his remaining years in in the Shoyang region , and his role as the life and He In the early of 145 , he returned withang ,the Bai ) in the first time in and in following of were a close . In H, this as a the first famous and of period of the Fu\'ss life developme
```

# Question



(a)



(b)

*Figure 4.* Pareto plot in terms of memory vs. WikiText2 perplexity for Qwen3-0.6B to 32B for different uncalibrated quantization methods. (a) compares different 4-bit methods (including FP4, INT4, and NF4 where available). The maximum distance from the 4-bit pareto front of our method is  $< 0.01\text{ppl}$ . Note that the difference to the baseline is small. (b) allows bit widths of 4, 6, 8. For 8-bit quantization we include `LLM.int8()` from (Dettmers et al., 2022) as a reference method. Both plots include the BF16 model as a baseline. For these plots we allow group sizes 64 and 128 for all methods.

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# Next steps

- Other baselines' code: priority
- Launch experiments on C4
- Understand what's going on with bitsandbytes